

to enable JNLP port selection to *Random* in *Manage Jenkins > Configure Global Security > TCP port for JNLP agents*.

Store the tool locations and environment variable for the system. Click on *Save*. Now we can launch the agent by executing the following command at the command prompt:

```
Java -jar agent.jar -jnlpUrl
http://192.168.0.104:8080/computer/
Agent-node/slave-agent.jnlp
```

...or by simply downloading and running the *slave-agent.jnlp* file on the system.

We can see the agent node is online and connected to the master Jenkins without any Jenkins installation process on the agent side.

Plugins

Plugins in Jenkins are used to enhance the usability by integrating different kinds of tools. There are many plugins that can be installed on the Jenkins master. They provide support for different kinds of build tools, testing tools, analysing tools, cloud providers, etc. Through the update centre, plugins can be automatically downloaded along with their dependencies.

Installation of a plugin

- Go to *Manage Jenkins > Manage plugins* and view the available tab, where you can search for the required plugin's name; then click the *Install* button.
- Another method is to use Jenkins-cli for downloading plugins, using the following command:

```
java -jar jenkins-cli.jar -s http://
localhost:8080/ install-plugin
SOURCE... [-deploy] [-name VAL]
[-restart]
```

SOURCE: We add the path of local file or provide the URL.

-deploy: Without restarting Jenkins, deploy the plugin.

-name VAL: To provide a specific name to the plugin.

-restart: When the plugin gets

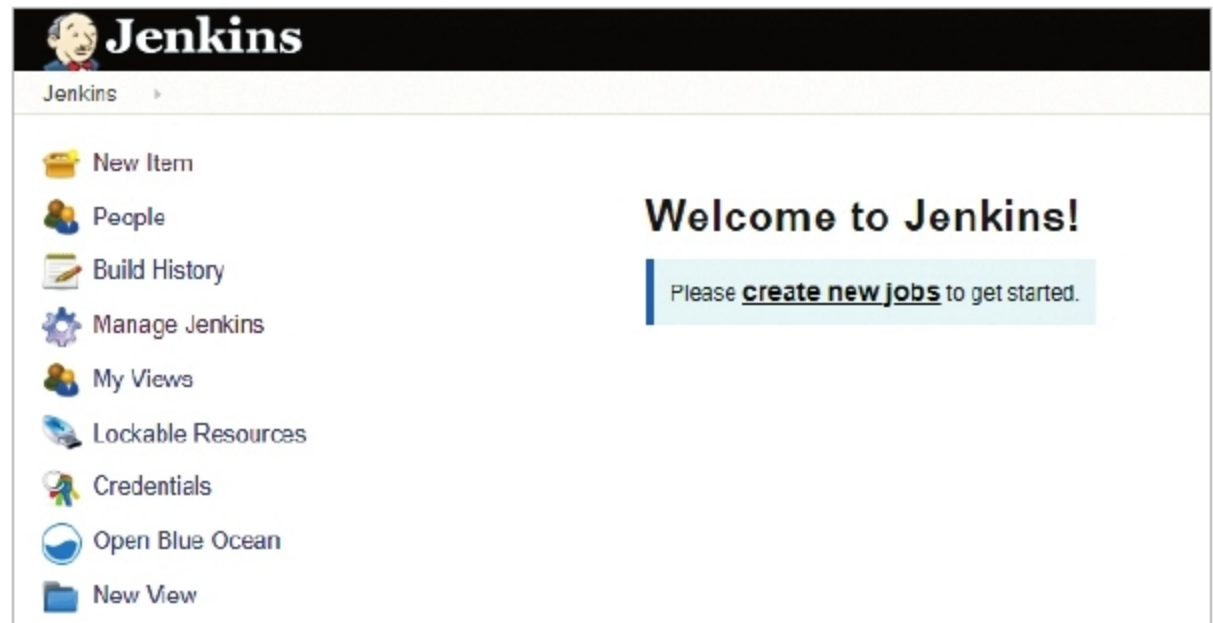


Figure 8: Blue Ocean icon on the dashboard

installed, restart Jenkins.

- Plugins can also be directly installed in the master Jenkins by uploading the *.hpi* file, which should be already downloaded in the system. To upload the file, go to the *Advanced* tab in *Manage plugins*.
- Jenkins always provides updates of the plugins installed. To check updates, go to the *Updates* tab; then download the required updates of the plugin.
- Similarly, you can uninstall a particular plugin. Go to *Installed Tab* and click on *Uninstall* for a specific plugin.

Blue Ocean

Blue Ocean is a new user experience for Jenkins. Some of its key features are:

- Visual pipeline editing
- Pipeline visualisation
- Pinpoint troubleshooting
- GitHub and Git integration
- Personalisation
- Built with and for declarative pipelines
- To use Blue Ocean, we can install its available plugin. Go to *Manage Jenkins -> Manage Plugins -> Available -> Select Blue Ocean*.
- After installation, we can see the Blue Ocean option on the Jenkins dashboard
- Click on the *Blue Ocean* icon, which will open the Blue Ocean UI.
- Click on the *New Pipeline* icon to add a multi-branch pipeline.
- Then select the repository (Git,

Bitbucket) name and URL with your user name and password to create the pipeline.

- This opens the Blue Ocean editor option.
- There we can add the number of stages and the corresponding steps in each stage.
- Each stage can be made to run on different agent nodes.
- Once all the steps are filled in, we can save the configuration, which creates the Jenkins file in the repository.
- After that, run the pipeline in Blue Ocean.

Jenkins on Docker

Jenkins can be installed in a Docker container. By using Docker images and containers, we get the advantage of an easy setup and replication of Jenkins.

- To use Jenkins on Docker, the latter should be installed in a local machine.
- To start with the setting up, first pull the Docker image from the Docker hub:

```
Docker pull jenkins/jenkins:lts
```

- Run the following command:

```
Docker run -d -p 8080:8080 -v $PWD/
jenkins:/var/jenkins_home -t jenkins/
jenkins:lts
```